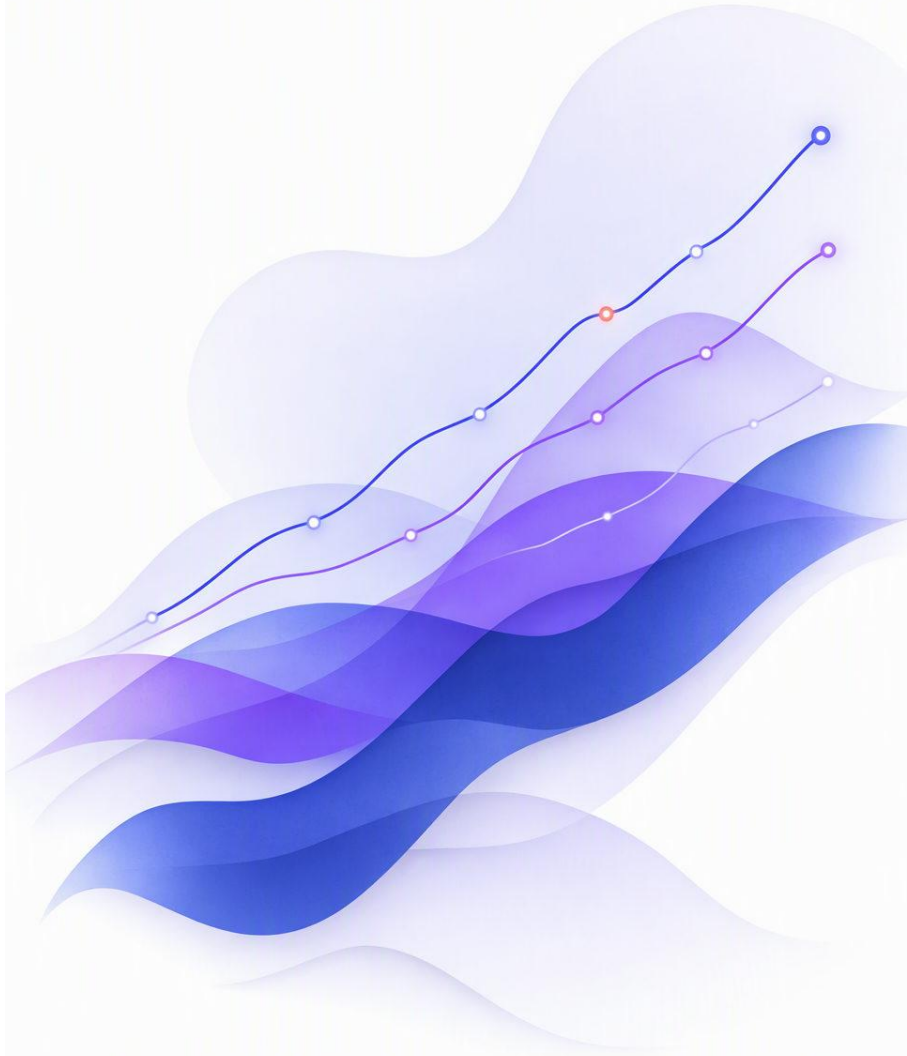
PRODUCT MANAGEMENT 101 · SESSION 7

Analytics for Product Strategy

How do you measure what matters in product performance?

Aditya Turalapati
Product Manager, Microsoft



SESSION 7

Which metric best represents user success?

"Which metric best represents user success?"

Pick one product you use daily. Name the one number.

THE LITMUS TEST

If this number doubles, is the user genuinely better off?

We'll sort your answers live

Activity metrics
Time spent, clicks, sessions — easy to count, easy to game
e.g., minutes in app

Outcome metrics
The job got done: a lesson learned, a ride taken, a payment cleared
e.g., rides completed per week

Business metrics
Revenue and renewals — they lag the user's success
e.g., monthly recurring revenue

SESSION 7 · AGENDA

Why metrics matter
Strategy → metric, vanity vs actionable, Goodhart

Anatomy of a metric
Metric spec · leading/lagging · inputs · guardrails

North Star metrics
Criteria · sector examples · trees · 2 cases

Funnels and AARRR
Cases: Airbnb, Expedia, Dropbox, Duolingo

HEART, sectors and root-cause analysis
HEART · sector cheat-sheet · RCA: Stripe, Meta

Experiments, Stripe capstone, wrap-up
A/B tests · Stripe war room · leaky bucket

VANITY OR ACTIONABLE?

· WRITE A METRIC SPEC

· FIND THE NORTH STAR

· FIX THE LEAK
· AARRR FOR A STARTUP

· HEART A FEATURE
· RCA SCENARIOS

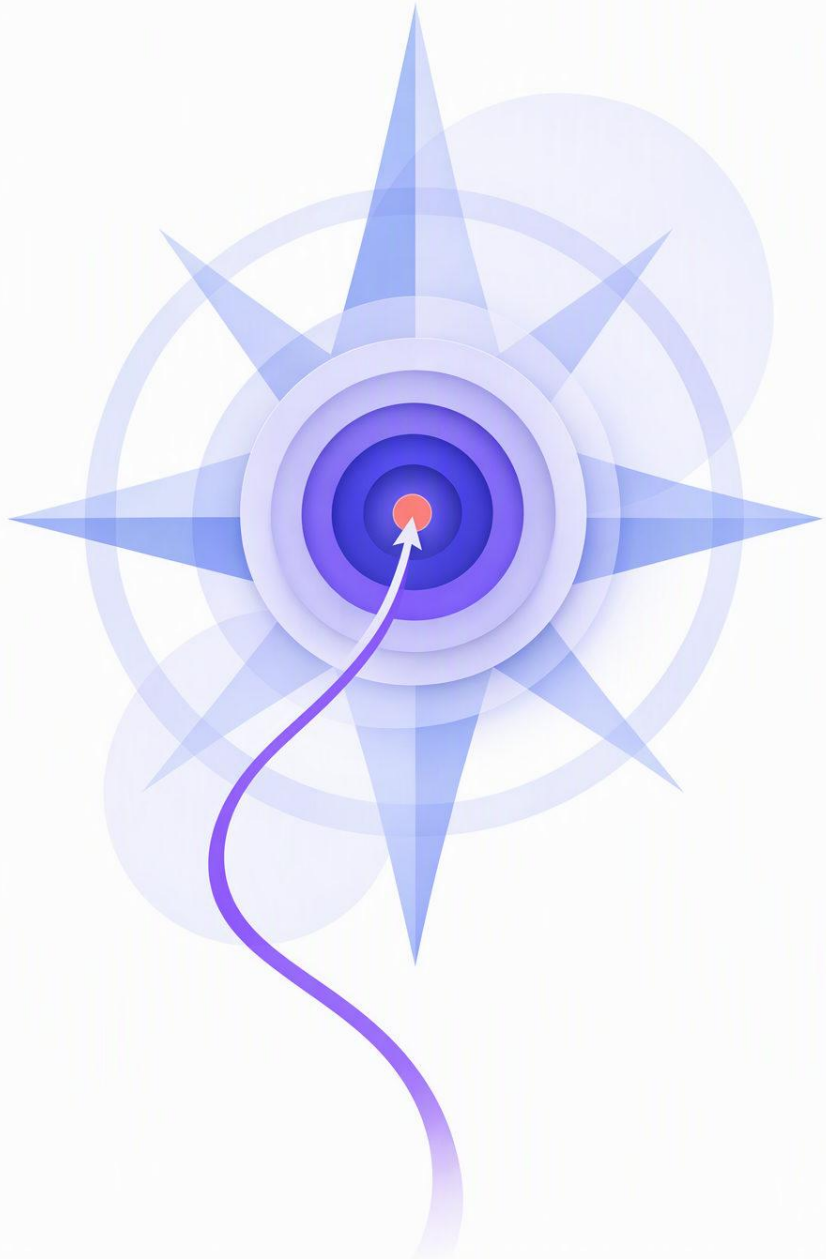
· STRIPE WAR ROOM

MODULE 1 OF 6

01

Why metrics matter

Connecting metrics to strategic intent — and what happens when they drift apart



SESSION 7 · WHY METRICS MATTER

Metrics turn strategy into decisions

Example	Strategic intent	Objective	Metric that tracks it
Duolingo B2C	Make learning a daily habit, for free	Bring more learners back every single day	**Daily active users; streak retention**
Shopify B2B	Help anyone start and grow a business	More merchants selling successfully	**Gross merchandise volume (GMV)**
Netflix B2C	Be the first choice for entertainment	Members find shows they love, so they stay	**Viewing per member; retention**

PM INSIGHT · WHAT TO STEAL

If you can't trace a metric back to a strategic bet, it's just a number. Start from intent, then choose what to count.

SESSION 7 · WHY METRICS MATTER

Vanity metrics flatter; actionable ones decide

Vanity

Goes up and to the right; changes no decision

- Total registered users
- App downloads
- Page views
- Cumulative revenue
- Social media followers

Actionable

Ratios, cohorts and rates that tell you what to do

- % of sign-ups activated within 7 days
- Week-4 retention, by cohort
- Conversion at each funnel step
- Revenue per active customer
- Net revenue retention (B2B)

PM INSIGHT · WHAT TO STEAL

Ask: “If this number moved 10% tomorrow, would we do anything differently?” If not, it's vanity. Eric Ries popularised the term in The Lean Startup.

SESSION 7 · WHY METRICS MATTER

Goodhart's law: targets bend behaviour

"When a measure becomes a target, it ceases to be a good measure."

Goodhart's law, as popularly phrased by anthropologist Marilyn Strathern

FAILURE

Wells Fargo · cross-selling

Sales targets for "products per customer" led staff to open accounts customers never asked for (the 2016 scandal).

Steal: a target with no customer-consent guardrail invites gaming.

COURSE-CORRECTION

Engagement-ranked feeds

Ranking feeds for clicks and engagement rewarded clickbait. Facebook publicly re-weighted News Feed in 2018 toward "meaningful social interactions".

Steal: engagement is a proxy; check it still means value.

SUCCESS

YouTube · views → watch time

In 2012 YouTube shifted recommendations from views toward watch time, to stop rewarding misleading thumbnails.

Steal: change the metric when it stops reflecting value.

Rule of thumb: pair every target with a second metric that would expose gaming.

HANDS-ON 1

Vanity or actionable? Rapid sort

Sort each metric: vanity or actionable? Then rewrite three 'vanity' ones into actionable versions.

GROUP
Pairs

HOW
Write V or A beside each — no laptops

DELIVERABLE
Sorted list + 3 rewrites

DEBRIEF
Which one was hardest to classify — and why?

The 10 metrics

1. Total app downloads
2. % of sign-ups onboarded within 7 days
3. Monthly page views
4. Week-4 retention of the March cohort
5. Registered users (all-time)
6. Mobile checkout conversion rate
7. Social media followers
8. Net revenue retention (NRR)
9. Total hours watched (all-time)
10. Time to first value for new teams

MODULE 2 OF 6

02

Anatomy of a metric

Definitions, windows, owners, leading vs lagging, inputs vs outputs and guardrails



SESSION 7 · ANATOMY OF A METRIC

A metric is a precise contract

Field	The question it answers	Example: week-1 activation
Name	What do we call it?	Week-1 activation rate
Definition	What exactly counts?	3+ invites and 1 core action
Formula	How is it computed?	Activated ÷ new workspaces
Unit	What is it measured in?	% of the sign-up cohort
Time window	Over what period?	Days 0–7; reported weekly
Segment	For whom?	Plan, region, channel
Data source	Where does it come from?	Product event logs
Owner	Who answers for it?	Growth PM
Target	What does good look like?	Baseline +5 pts per quarter
Guardrail	What must not break?	Support tickets per workspace

PM INSIGHT

Test the spec: could a new analyst rebuild the exact number from these ten lines alone?

MOST FIGHTS START HERE

Definition and time window

SESSION 7 · ANATOMY OF A METRIC

Manage the inputs, watch the outputs

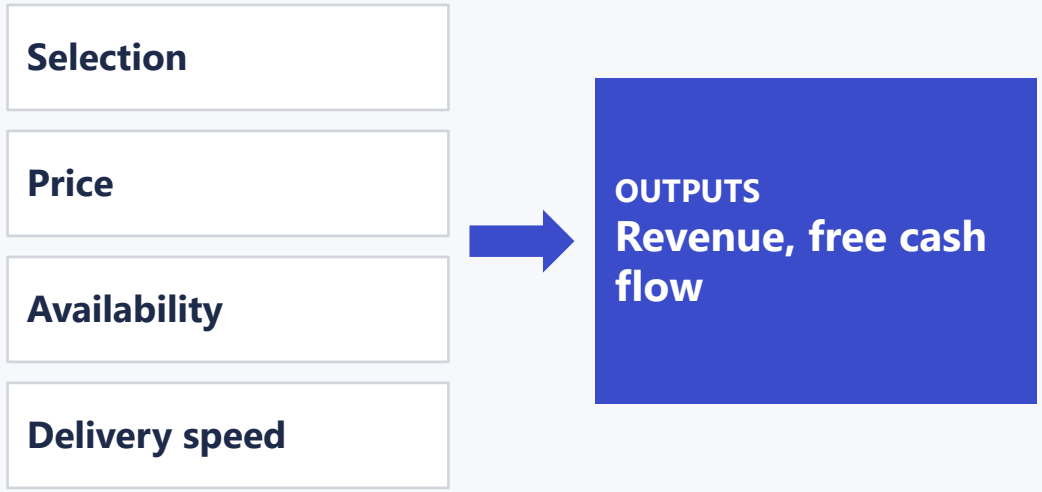
Leading vs lagging

LEADING
Moves first; you can act on it

LAGGING
Confirms later; hard to steer

Slack: teams that crossed an early messaging threshold (a commonly cited ~2,000 messages) tended to stick — a leading signal of retention.
Subscriptions: trial activation leads; renewal revenue lags.
B2B SaaS: weekly active seats lead; churn and NRR lag.

Input vs output · Amazon



Amazon reviews controllable inputs every week; outputs follow (Working Backwards, 2021).

PM INSIGHT
You cannot ship an output. Pick inputs your team can move this sprint.

SESSION 7 · ANATOMY OF A METRIC

Every target needs a guardrail

Product	Push this…	…without breaking this	Why
Ride-hailing e.g., Uber	**Pickup time (ETA)**	**Driver earnings & cancellations**	Faster pickups bought with idle drivers burn the other side of the marketplace.
Streaming e.g., Netflix	**Hours viewed**	**Cancellations & satisfaction**	Autoplay can inflate hours without making members happier.
Support desk e.g., any SaaS	**Tickets closed per hour**	**Reopen rate & CSAT**	Closing fast is not the same as solving.
Growth email e.g., any B2C app	**Click-through rate**	**Unsubscribes & spam reports**	Every extra email borrows trust from the next one.

PM INSIGHT
Microsoft, Booking.com and Netflix define guardrail metrics that can veto a “winning” test.

HANDS-ON 2

Write a metric spec for a real feature

In pairs: pick one feature and define its success metric so precisely that any analyst would compute the same number.

PICK ONE FEATURE

Spotify Blend
A shared playlist for two friends

Uber ride-share
Strangers share a cheaper ride

Notion AI
Writing help inside your docs

PRODUCE
One spec card: all 8 fields, incl. a guardrail

DEBRIEF
Could two analysts compute the same number?

YOUR METRIC SPEC

NAME
What do we call it?

DEFINITION
What exactly counts?

FORMULA
Numerator ÷ denominator

TIME WINDOW
Over what period?

SEGMENT
For whom? Cut by what?

TARGET
Baseline → goal, by when

OWNER
Who answers for it?

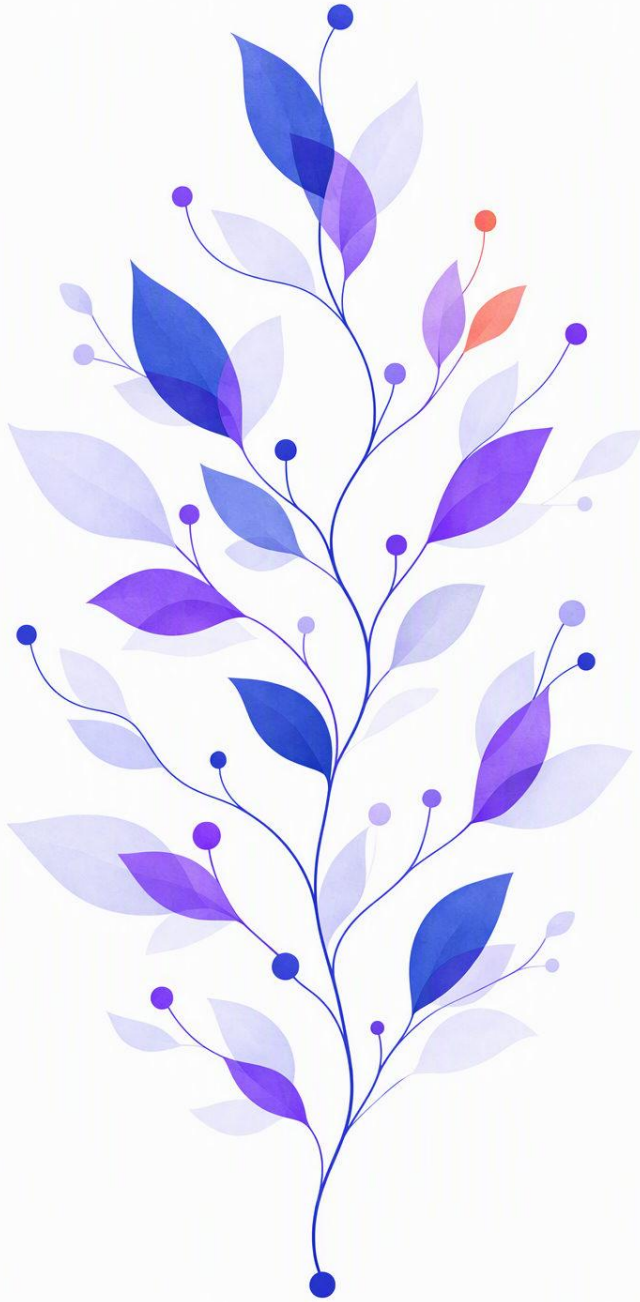
GUARDRAIL
What must not break?

MODULE 3 OF 6

03

North Star metrics

One metric for core value, the tree behind it, and two that went wrong



SESSION 7 · NORTH STAR METRICS

North Star: one metric for core value

The single metric that best captures the value customers get — and predicts long-term business results.

1. **Reflects customer value**
2. **Predicts revenue (leading)**
3. **Actionable by teams**
4. **Easy to understand**

WHERE IT SITS

CUSTOMER VALUE
A guest has a great stay; a host earns

→

NORTH STAR
Nights booked
Airbnb · commonly cited

→

BUSINESS RESULTS
Revenue and profit follow — but lag

PM INSIGHT
Revenue is rarely a good North Star: it lags, and a price rise can lift it while users suffer.

Term popularised by Sean Ellis and Amplitude. Airbnb's "nights booked" is a commonly cited example.

SESSION 7 · NORTH STAR METRICS

Every sector has its own North Star

MARKETPLACE
Airbnb
Nights booked

B2B SAAS
Slack
Messages sent within teams

FINTECH / PAYMENTS
Stripe
Total payment volume

MEDIA / STREAMING
Spotify
Time spent listening

SOCIAL
Facebook
Daily active users

EDTECH
Duolingo
Daily active users

E-COMMERCE
e.g., Amazon, Flipkart
Orders per active customer

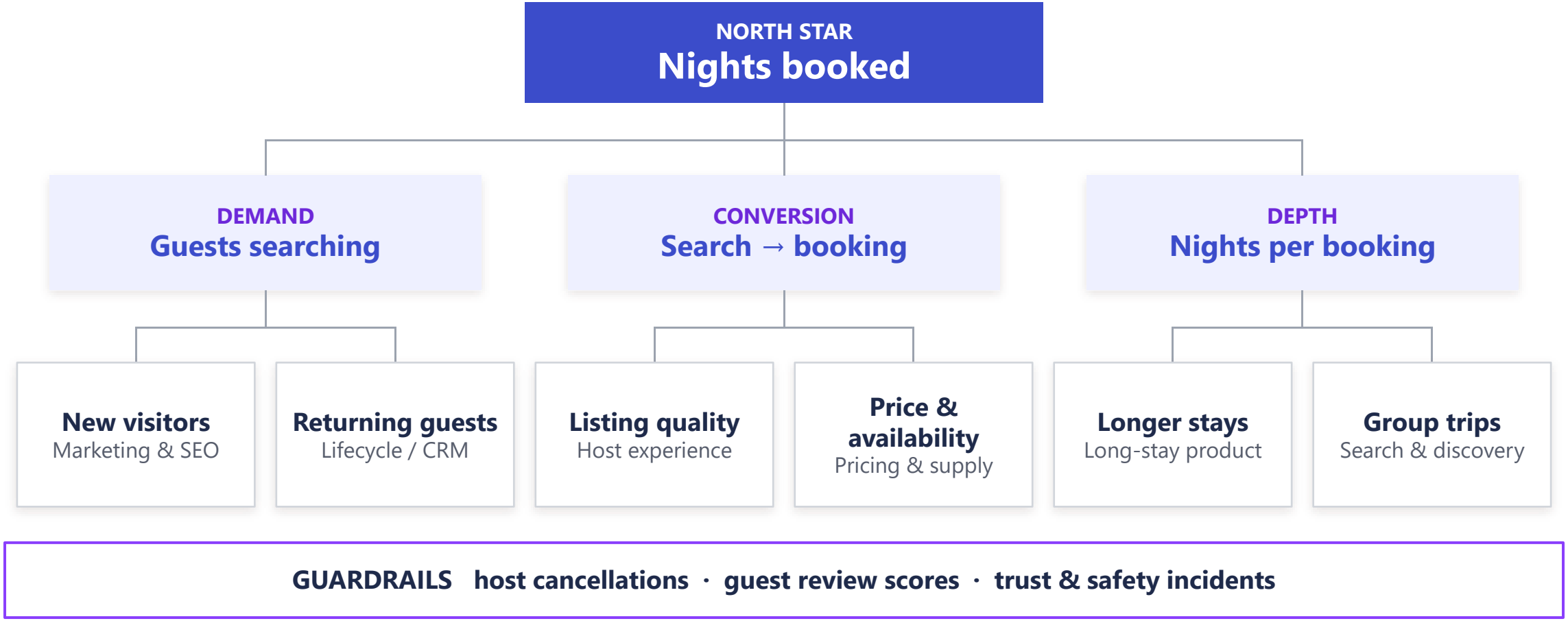
HEALTH
e.g., Practo
Consultations completed

GAMING
Roblox
Hours engaged

Commonly cited examples from public talks, filings and articles; the e-commerce and health picks are illustrative. Internal metrics vary and change.

Company	Category	Bad NSM	Actual NSM	Game
facebook	Social media	Daily posts	Daily active users	Attention
Spotify	SaaS, music	The number of Subscribers	Time spent listening to music	Attention
amazon	E-commerce	Monthly Revenue	The number of orders per month	Transaction
Uber	Rides	The number of installs	The number of rides per week	Transaction
loom	SaaS, video sharing	LTV/CAC	Videos with a view	Productivity
airbnb	Booking platform	Time spent using the app	Nights booked	Transaction
tinder	Dating app	The number of swipes right	Matches made	Transaction
fitbit	Wearable fitness products	The number of products sold	Steps made	Productivity
Dropbox	File hosting	The number of paid customers	Files saved	Productivity
Canva	Graphic design	The number of projects created	Happy, active users	Productivity
YouTube	Video	The number of videos uploaded	Time spent watching videos	Attention
WhatsApp	Messaging	The number of accounts	Messages send	Productivity
Quora	Q&A knowledge sharing	Time spent in the app	The number of answers	Productivity
NETFLIX	SaaS, video streaming	The number of subscribers	Weekly viewing hours	Attention

Trees turn a North Star into team goals



Illustrative decomposition for teaching — not Airbnb's internal tree.

CASE STUDY · WRONG NORTH STAR

YouTube: from views to watch time

WHAT THEY MEASURED

Views (clicks)

Discovery leaned on how often a video was clicked and started.

BEHAVIOUR IT CAUSED

Clickbait won

Misleading titles and thumbnails earned clicks; viewers left within seconds, feeling tricked.

WHAT THEY CHANGED

Watch time (2012)

Search and recommendations began favouring videos people actually kept watching.

INSIGHT

A click measures curiosity, not value. Pick the metric that only grows when the viewer is satisfied — and expect creators (and teams) to optimise whatever you choose.

Public source: YouTube Creator Blog announcement, 2012; widely reported. No figures shown.

CASE STUDY · WRONG NORTH STAR

Facebook: from time spent to real connection

WHAT THEY MEASURED

Engagement & time

News Feed ranking rewarded posts that held attention and drew clicks.

BEHAVIOUR IT CAUSED

Passive scrolling

Viral video and publisher posts crowded out friends and family; consumption became passive.

WHAT THEY CHANGED

Meaningful interactions

Jan 2018: ranking would favour posts that spark conversations between people.

INSIGHT

Leadership accepted a lower engagement number to protect long-term value. Changing a North Star is a strategy call: announce it, and say what you expect to fall.

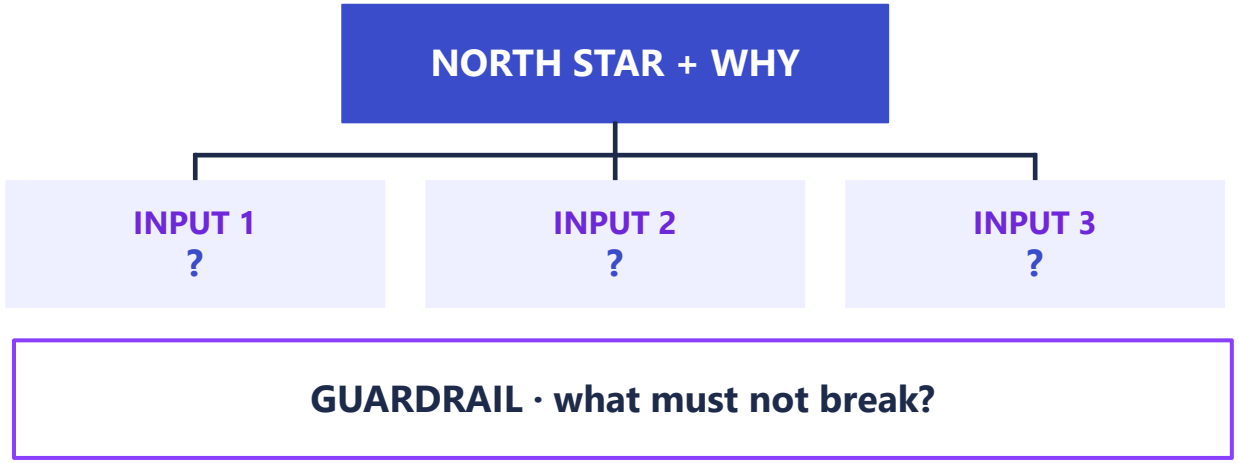
Public source: Mark Zuckerberg's Facebook post, January 2018; widely reported. No figures shown.

HANDS-ON 3 · ROUND 1

Find the North Star: sector rounds

Each team draws three products from different sectors. Find each one's North Star, then sketch its first metric tree.

Zomato / DoorDash Food delivery	**Salesforce** B2B CRM	**Duolingo** Edtech	**Netflix** Streaming
Razorpay / Stripe Payments	**Practo** Healthtech	**Canva** Design SaaS	**Tinder** Dating app



GROUP Teams of 2 · Pick 2 products per team

PRODUCE
Per product: NSM + why + 3 inputs + 1 guardrail

DEBRIEF
Which sector was hardest - and why?

HANDS-ON 3 · ROUND 2

Spot the bad North Star

Each proposed North Star below is flawed. Say what behaviour it would reward, then propose a better one.

Meditation app
e.g., Headspace
App downloads

What would it reward?

Better North Star?

Dating app
e.g., Tinder
Swipes per day

What would it reward?

Better North Star?

B2B CRM
e.g., Salesforce
Licences sold

What would it reward?

Better North Star?

Online courses
e.g., Coursera
Courses enrolled

What would it reward?

Better North Star?

PRODUCE
Per card: the flaw in one line + a better metric

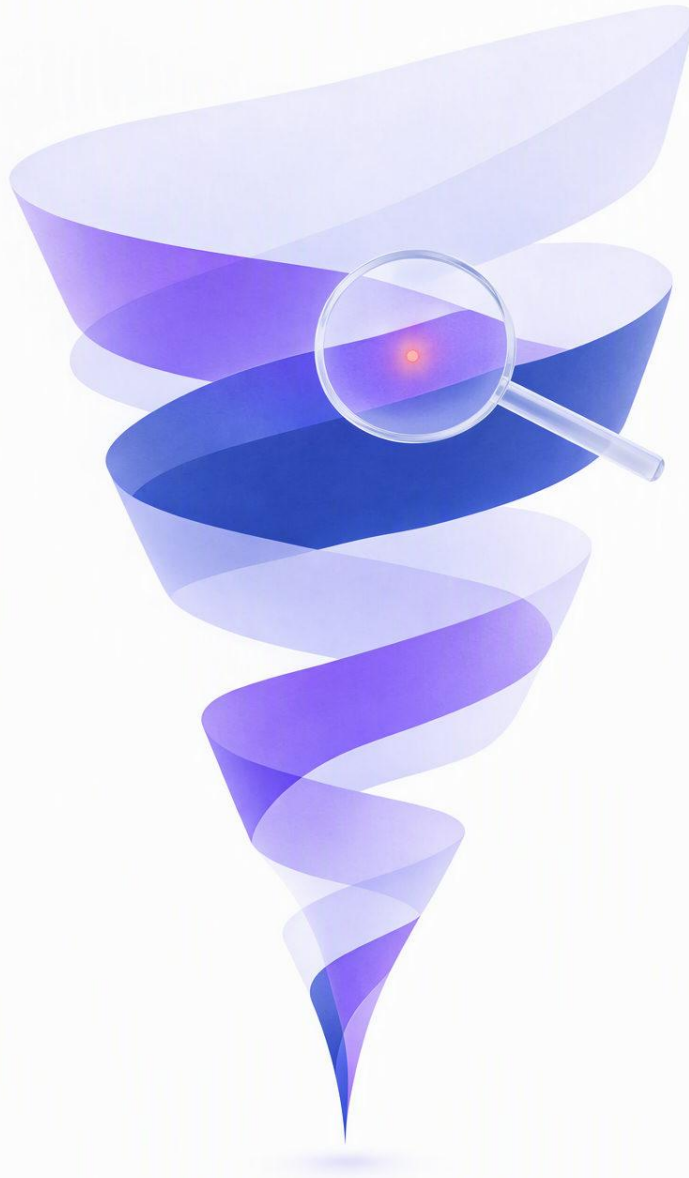
DEBRIEF
Which bad one looks most like a real dashboard?

MODULE 4 OF 6

04

Customer-journey metrics

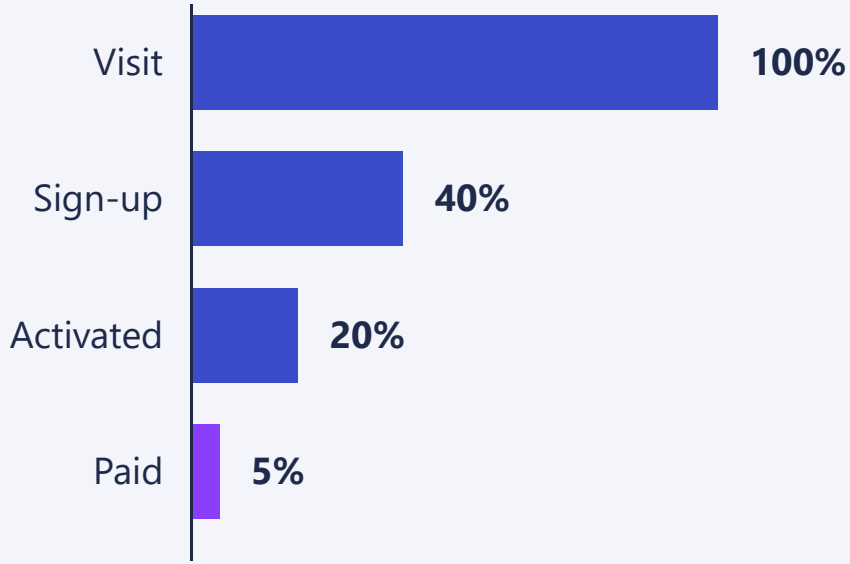
Funnels, AARRR and HEART — each with real cases and an exercise



SESSION 7 · FUNNELS

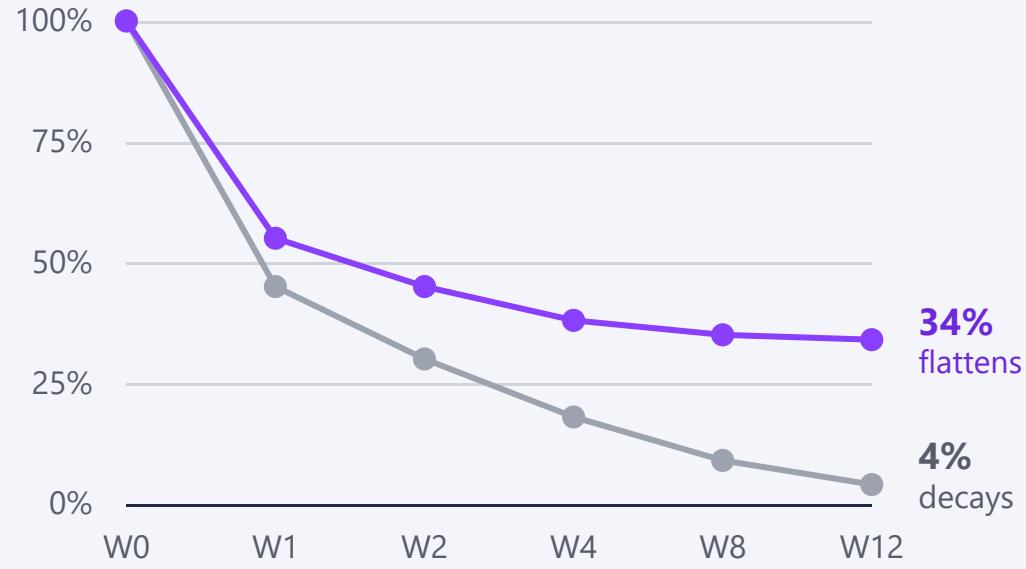
Funnels show where; cohorts show whether

Conversion funnel



Biggest step loss: activated → paid (75% drop). Illustrative numbers.

Cohort retention curves



% of a sign-up cohort still active, by week. Illustrative. A flattening curve is a product-market-fit signal.

CASE STUDY · FUNNELS

Airbnb: better photos, more bookings

Search → **View listing** → **Book** → Stay & review

WHAT THEY SAW

Viewed, not booked

Early New York listings got views, but too few turned into bookings.

THE DIAGNOSIS

Poor photos

Visiting hosts, the founders saw amateur phone photos making good homes look unappealing.

WHAT THEY CHANGED

Pro photography

They shot listings themselves, later a free photo programme; bookings rose.

PM INSIGHT

The funnel told them WHERE (view → book); talking to hosts told them WHY. Do things that don't scale to find the fix — then scale it.

Widely reported founder story (Joe Gebbia, Paul Graham). Figures vary by telling, so none are shown here.

CASE STUDY · FUNNELS

Expedia: one form field, ~$12M a year

Search → Pick a hotel → Enter details → **Pay** → Booked

WHAT THEY SAW

Drop-off at payment

Many customers clicked to buy but did not complete the booking.

THE DIAGNOSIS

A confusing field

An optional "Company" box was misread; people typed their bank's name and the card address check failed.

WHAT THEY CHANGED

Field deleted

Removing the field is widely reported to have recovered ~$12M a year in profit.

PM INSIGHT

The most valuable fix can be a deletion. Instrument every step — and every field — so a one-box leak is visible.

Widely reported (2010) from Expedia's then analytics leadership; the ~$12M annual profit is commonly cited, so treat it as approximate.

HANDS-ON 4

Fix the leak: a D2C skincare funnel



YOUR TASK
Teams of 4: find the biggest leak and list 3 likely causes

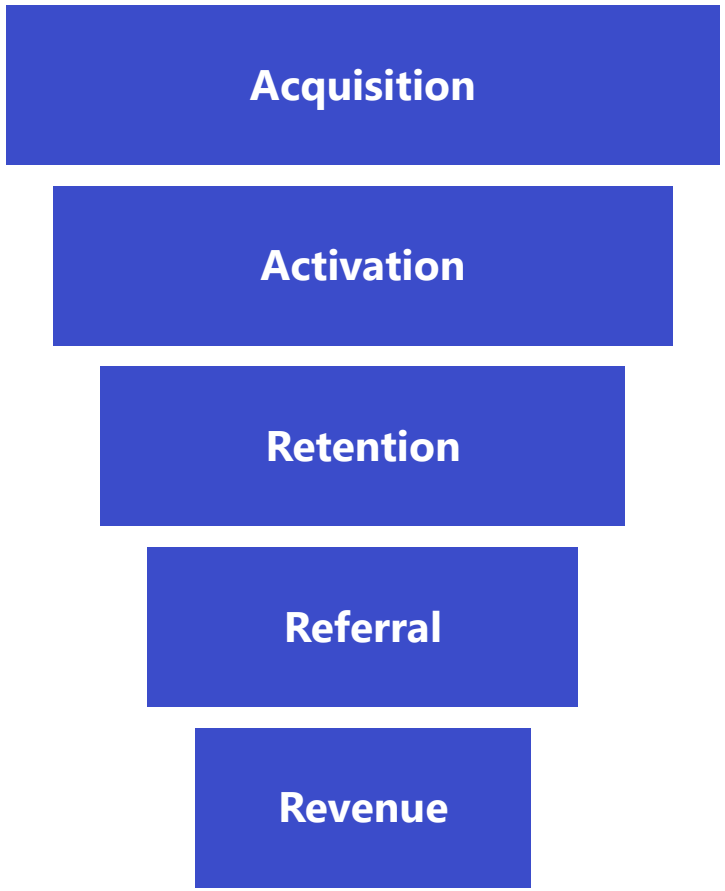
PRODUCE
One experiment: the change, its success metric, its guardrail

DEBRIEF
Fix the biggest drop — or the cheapest one first?

Illustrative numbers for a fictional brand. Step % = this step ÷ the step before. “Bought again” = within 60 days.

SESSION 7 · CUSTOMER-JOURNEY METRICS

AARRR: follow the customer journey



	Question	B2C · e.g., Duolingo	B2B · e.g., Slack
Acquisition	**How do people find us?**	App-store installs by channel	Sign-ups from invites & search
Activation	**Is the first experience great?**	First lesson done on day 1	Team sends messages in week 1
Retention	**Do they come back?**	Day-7 and day-30 return rate	Active teams after 8 weeks
Referral	**Do they tell others?**	Friend invites sent	New members invited per team
Revenue	**Do they pay?**	Free-to-paid conversion	Paid seats; expansion revenue

CASE STUDY · AARRR

Dropbox: referrals powered acquisition

Acquisition | Activation | Retention | **Referral** | Revenue

THE PROBLEM

Paid ads didn't pay

Early search ads reportedly cost far more per customer than a free user was worth.

THE LOOP

Give space, get space

Invite a friend and both of you earn extra free storage — the reward is the product.

THE RESULT

Viral sign-ups

Widely cited as a major driver of Dropbox's rapid growth after launch.

PM INSIGHT

The best referral reward strengthens the core product. Measure the loop: invites sent per user × invite acceptance rate.

Public source: Drew Houston, Startup Lessons Learned conference, 2010; widely cited. Figures kept qualitative.

CASE STUDY · AARRR

Duolingo: growth through retention

Acquisition	Activation	**Retention**	Referral	Revenue

THE INSIGHT

Retention moves DAU

Modelling users by state showed current-user retention had the biggest lever on daily actives.

THE LEVERS

Streaks & leagues

Streaks, streak protection and weekly leaderboard leagues give a reason to return daily.

THE RESULT

Growth re-accelerated

Duolingo publicly credits this retention focus with reigniting daily-active-user growth.

PM INSIGHT

Model your users by state (new, current, at-risk, dormant) and find the transition that moves your North Star most — often keeping users, not winning new ones.

Public source: Duolingo's own writing (e.g., former CPO Jorge Mazal, Lenny's Newsletter, 2022); widely cited. Kept qualitative.

HANDS-ON 5

AARRR for a startup

Pick one company. Give one metric per stage, find the weakest stage, and design one experiment to fix it.

Acquisition	Metric: ?	Weakest?
Activation	Metric: ?	Weakest?
Retention	Metric: ?	Weakest?
Referral	Metric: ?	Weakest?
Revenue	Metric: ?	Weakest?

ONE EXPERIMENT FOR THE WEAKEST STAGE
Change → success metric → guardrail

PICK ONE COMPANY

Zepto / Blinkit	**Notion**
CRED	**Headspace**
Freshworks	**Your own pick**

PRODUCE
5 metrics · weakest stage · 1 experiment

DEBRIEF
Does the weakest stage change for a B2B company?

SESSION 7 · CUSTOMER-JOURNEY METRICS

Google's HEART: measure the experience

H

Happiness
How users feel: satisfaction, ease
e.g., in-product survey score

E

Engagement
Depth and frequency of use
e.g., actions per user per week

A

Adoption
New users or new-feature uptake
e.g., % of accounts using a feature

R

Retention
Users who keep coming back
e.g., % active again after 30 days

T

Task success
Efficiency, effectiveness, errors
e.g., time to complete checkout

GOALS
What does success look like?

SIGNALS
Which behaviour shows it?

METRICS
How do we count that signal?

PM INSIGHT · WHAT TO STEAL
Pick only the dimensions that matter: a payments flow lives on task success; a social feed on engagement and happiness.

HEART in action: Spotify Discover Weekly

HEART	Goal	Signal	Metric
H · Happiness	It “gets” my taste	Saves, likes, survey	% rating it 4–5 of 5
E · Engagement	People listen deeply	Tracks played past 30 s	Tracks streamed per listener
A · Adoption	New users find it	First play of the list	% of weekly users who open it
R · Retention	A Monday habit	Returns week after week	% back 4 weeks in a row
T · Task success	Find new music fast	Skips; saves to library	Skip rate (guardrail)

PM INSIGHT
Start from the goal, not the dashboard. You rarely need all five letters — pick the two or three that match the feature’s job.

Illustrative application of Google’s HEART framework (Rodden, Hutchinson & Fu, 2010) — not Spotify’s internal metrics.

HANDS-ON 6

HEART a feature

Pick one feature. Choose the 2–3 HEART letters that matter most, then go Goal → Signal → Metric for each.

	GOAL	SIGNAL	METRIC
H Happiness	?	?	?
E Engagement	?	?	?
A Adoption	?	?	?
R Retention	?	?	?
T Task success	?	?	?

PICK ONE FEATURE

Smart Compose
Gmail

Group order
Uber Eats

Easy Apply
LinkedIn

PRODUCE
2–3 rows filled · one metric marked as guardrail

DEBRIEF
Which letter did you drop — and why?

SESSION 7 · METRICS BY SECTOR

B2B SaaS and B2C read different scoreboards

Question	B2B SaaS · e.g., Atlassian, Snowflake	B2C · e.g., Spotify, Duolingo
How big is the business?	MRR / ARR	ARPU × paying users
Are customers growing with us?	Net revenue retention; seat expansion	Upgrades to premium or family plans
Who is leaving?	Logo churn and revenue churn	Cohort retention: D1, D7, D30
Is growth efficient?	CAC, payback, LTV:CAC (≈3:1 rule of thumb)	Blended CAC vs LTV; organic share
Did they get value fast?	Activation; product-qualified leads	Onboarding done; first-session success
How often do they use it?	Weekly active seats per account	DAU/MAU stickiness
Where do they drop off?	Trial → paid conversion	Visit → sign-up → purchase

PM INSIGHT · WHAT TO STEAL

In B2B the buyer and the user differ: track account health (NRR) and user habit (active seats). Snowflake and Datadog are known for land-and-expand.

SESSION 7 · METRICS BY SECTOR

Metric cheat-sheet by sector

SAAS (B2B)
- ARR and NRR
- Weekly active seats
- Logo churn

MARKETPLACE
- GMV / bookings
- Match rate (liquidity)
- Take rate

FINTECH / PAYMENTS
- Payment volume
- Authorisation rate
- Fraud & disputes

E-COMMERCE
- Conversion rate
- Average order value
- Repeat-purchase rate

MEDIA / STREAMING
- Hours per user
- Paid subscribers
- Monthly churn

SOCIAL
- DAU / MAU
- Posts & messages per user
- New connections

EDTECH
- Lessons completed
- Course completion
- Learner D30 retention

HEALTHTECH
- Consults completed
- Repeat visits
- Patient satisfaction

GAMING
- D1 / D7 retention
- Session length
- ARPDAU

AI PRODUCTS
- Task success rate
- Weekly active users
- Cost per query

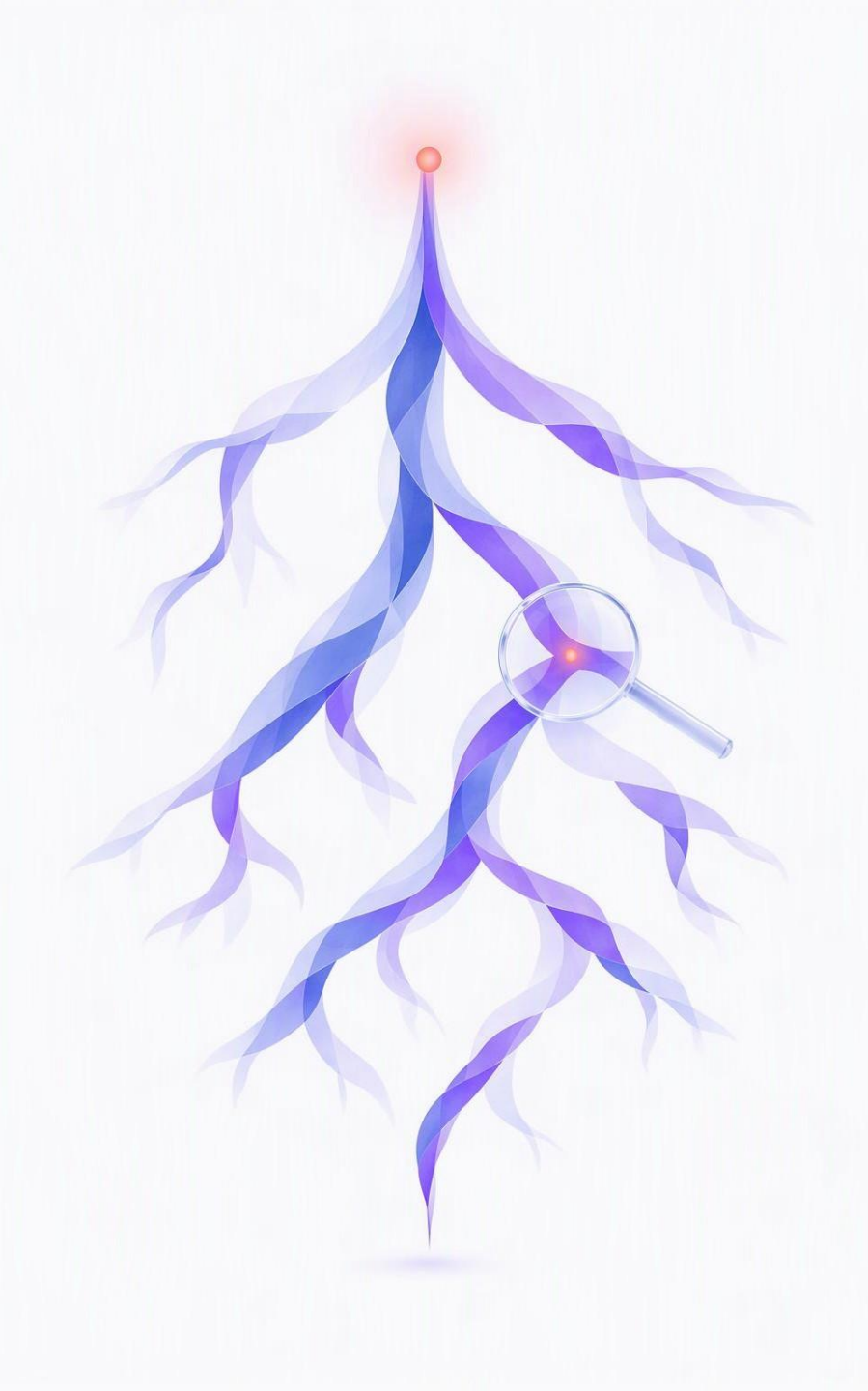
Commonly used metrics, not a complete list. Pick one North Star per product and pair it with a guardrail.

MODULE 5 OF 6

05

Root-cause analysis

When a metric moves: a structured sweep, two real cases, and A/B testing



SESSION 7 · ROOT-CAUSE ANALYSIS

“The metric dropped 10%.” Now what?

1. **Clarify & validate the data**
   Same definition? Tracking bug, bots, a logging change?
2. **Scope it in time**
   Start date? Sudden or gradual? One metric or many?
3. **Internal or external?**
   Our releases, tests, outages vs season, rivals, rules
4. **Segment it**
   Time, platform, geo, user type, channel, funnel step
5. **Hypothesise & prioritise**
   List causes; rank by likelihood × impact
6. **Validate & fix**
   Confirm with data or users; fix; watch a guardrail

WORKED EXAMPLE

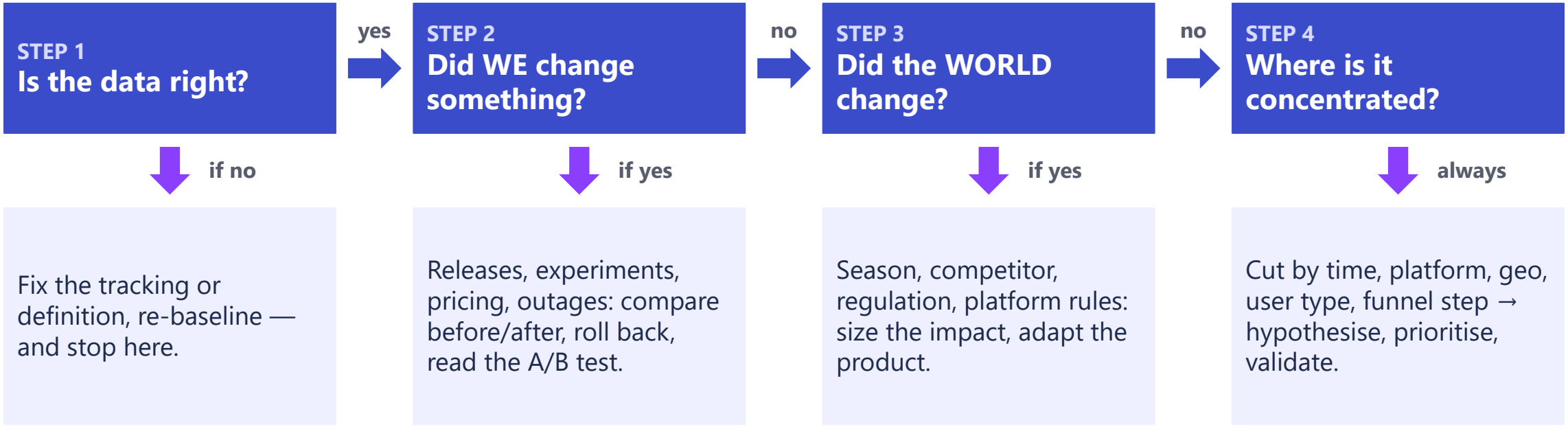
Rides in one city fall 10% on a Friday.
Real? Payments data agrees. Scope: Android only, since Tuesday’s release. External? No holiday or storm. Hypothesis: the new version breaks card payment on some phones.

PM INSIGHT

Interviewers grade the structure, not the answer. Say your plan out loud before you dig.

SESSION 7 · ROOT-CAUSE ANALYSIS

The RCA decision tree: cheap checks first



PM INSIGHT

Most drops are data bugs or our own releases — fastest to confirm, so check them first. External causes are rarer but bigger: next, Stripe and SCA, then Meta and iOS 14.5.

CASE STUDY · ROOT CAUSE

Stripe and SCA: a dip with an outside cause

THE SYMPTOM
Checkout success dips (illustrative: −6%)

SEGMENT · REGION
Only the EEA moves; elsewhere is flat

SEGMENT · CARD & STEP
Cards hitting 3D Secure drop off

ROOT CAUSE · EXTERNAL
EU PSD2 Strong Customer Authentication

STRIPE'S PUBLIC RESPONSE
- SCA-ready payment APIs that ask for authentication only when needed
- Support for 3D Secure 2: a lighter, in-app check
- Exemptions (e.g., low-value payments) applied where rules allow

PM INSIGHT
Segmentation turned a scary dip into one region, one card flow, one step — pointing to a known regulation. Check the regulatory calendar before blaming your release.

Teaching reconstruction grounded in a real event: PSD2 SCA took effect in the EEA from 14 Sept 2019, with phased enforcement. The dip figure is illustrative.

CASE STUDY · ROOT CAUSE

Meta and Apple's iOS 14.5 privacy change

THE SYMPTOM
Ad results and measured conversions weaken

SEGMENT · PLATFORM
Concentrated in iOS; Android far less hit

SEGMENT · TIMING
Starts with the iOS 14.5 rollout, 2021

ROOT CAUSE · EXTERNAL
App Tracking Transparency opt-outs

~$10B
Meta's own estimate of the 2022 revenue impact

META'S PUBLIC RESPONSE
- Aggregated Event Measurement for iOS
- Modelled conversions to fill signal gaps
- More AI-driven targeting and in-app actions

PM INSIGHT
When the cause is a platform you don't control, there is no rollback. The fix is a product response: rebuild measurement and depend less on the lost signal.

Sources: Apple iOS 14.5 release (April 2021); Meta Q4 2021 earnings call (Feb 2022) — the ~$10B estimate was widely reported.

HANDS-ON 7

RCA scenarios: rapid rounds

For each card: walk the RCA decision tree, list your top 3 hypotheses and name the first data cut you'd ask for.

ROUND 1 · FOOD DELIVERY

Swiggy / DoorDash orders down 12% week-on-week

ROUND 2 · B2B SAAS

Slack DAU flat, but messages per user down

ROUND 3 · STREAMING

Netflix sign-ups up, but 30-day retention down

ROUND 4 · PAYMENTS

Checkout success down in one country (Stripe-like)

GROUP
Teams of 4 · one card per round, then rotate

PRODUCE
Per card: tree walk · top 3 hypotheses · first data cut

DEBRIEF
Which first data cut ruled out the most?

Scenarios are fictional practice cases; company names set the context only.

SESSION 7 · EXPERIMENTS

A/B tests: let customers settle debates

Hypothesis
If we do X, metric Y moves, because Z.

Primary metric
One decision metric, chosen before launch.

Guardrails
Must not worsen: latency, revenue, complaints.

Sample size
Small effects need many users; fix size and duration upfront.

Significance
Is the gap bigger than chance? Don’t stop early.

Novelty effect
New things get clicks at first; run long enough.

EXPERIMENTATION CULTURE

Booking.com
Runs very many concurrent tests; most staff can launch one.

Netflix
Tests even the artwork shown for each title.

Microsoft
Bing’s platform: a small ad-headline tweak became a major revenue win.

PM INSIGHT · WHAT TO STEAL

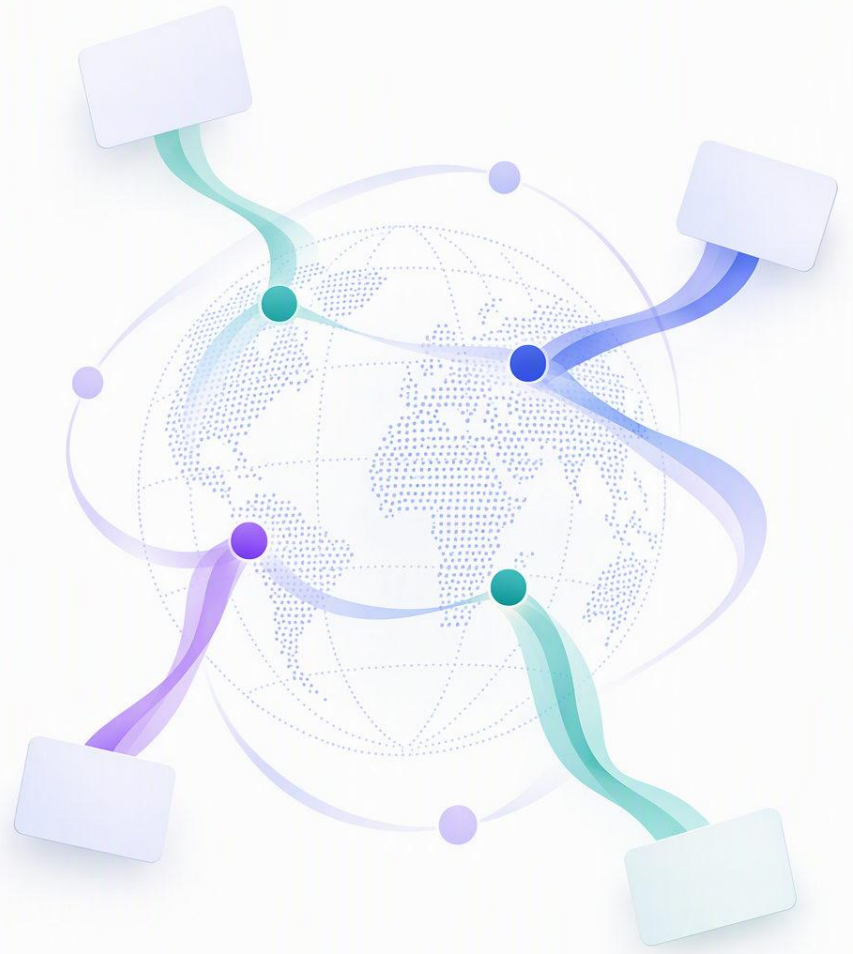
Most ideas don’t win. Experimentation leaders report that only a minority of tests improve the key metric — so test cheaply and often.

CASE STUDY · MODULE 6 OF 6

06

Stripe capstone

Asking the right questions about a B2B payments platform



CASE STUDY · STRIPE

Stripe: the internet's payments plumbing

THE BUSINESS

- Founded in 2010 by brothers Patrick and John Collison.
- Developer-first: clean APIs and docs let a few lines of code accept payments.
- Beyond Payments: Billing, Connect for marketplaces, Radar for fraud, Terminal.
- Serves startups to large enterprises worldwide.

Three users, three definitions of success

DEVELOPERS
Integrate fast; trust it in production
Success: time to first payment, clear errors, API uptime

FINANCE & OPS TEAMS
Money arrives, reconciles, stays safe
Success: authorisation rate, payout speed, low disputes

END CUSTOMERS (PAYERS)
Checkout just works, in their currency
Success: paid first time, local methods, no false declines

CASE STUDY · STRIPE

Stripe’s candidate North Star and drivers

CANDIDATES TO DEBATE

Total payment volume
Stripe publicly reports it each year — over $1 trillion in 2023

Successful payments
Counts value moments; ignores ticket size

Active merchants
Breadth of adoption; ignores depth



Teaching model built from public information — not Stripe’s internal metrics.

CASE STUDY · STRIPE

Guardrails keep volume growth honest

Fraud rate
Volume from fraud is a liability, not growth.

Dispute rate
Chargebacks hurt merchants and card-network standing.

API latency
A slow checkout loses the sale at the last step.

Uptime
Every minute down is lost revenue for every merchant.

PM INSIGHT
For infrastructure, reliability IS the product. A guardrail breach outranks any growth win.

THE SCENARIO

Payment volume keeps growing quarter on quarter, but…

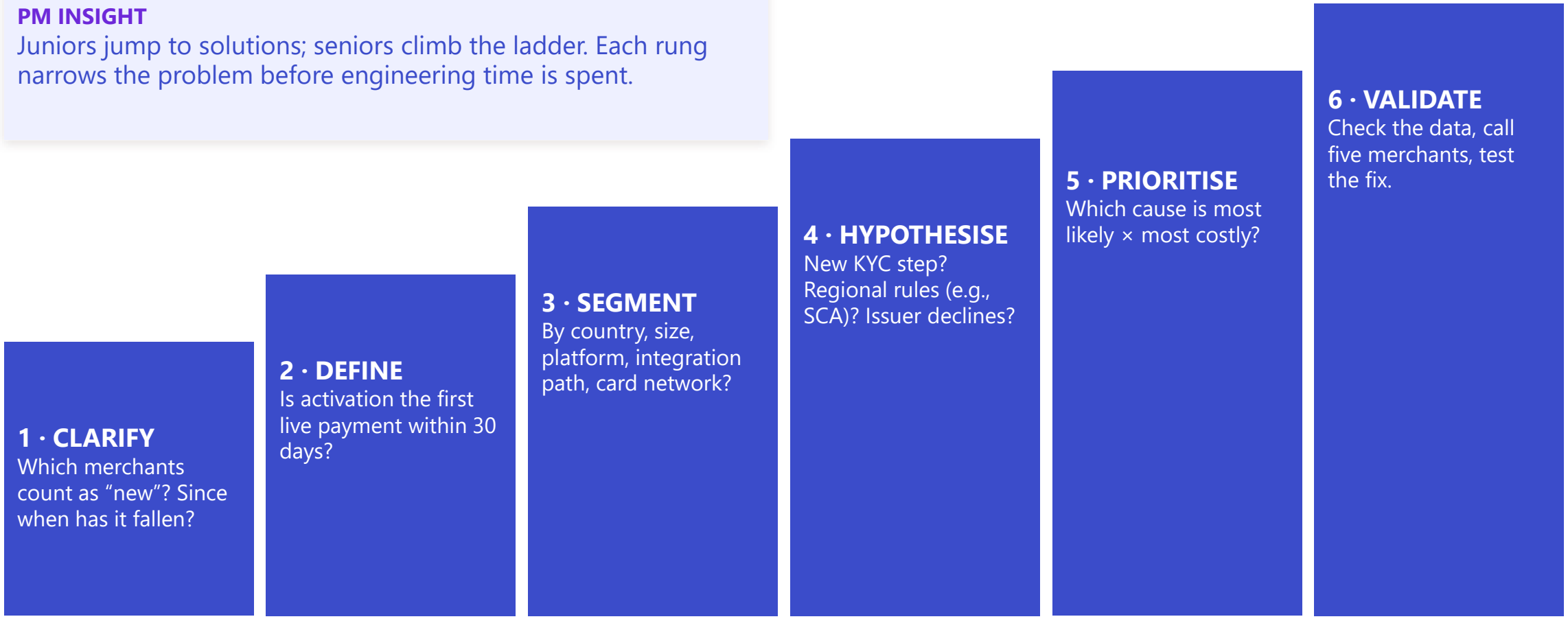
▲ Total payment volume: up
▼ New-merchant activation: falling for 3 months
▼ Authorisation rate: dipping in one region

Your job: decide what to measure and fix first.

CASE STUDY · STRIPE

The question ladder: climb, then conclude

PM INSIGHT
Juniors jump to solutions; seniors climb the ladder. Each rung narrows the problem before engineering time is spent.



HANDS-ON 8 · STRIPE WAR ROOM

CASE STUDY

Stripe war room: what do you measure first?

STEP 1
Clarify and define the metrics

STEP 2
Segment; list 4+ hypotheses

STEP 3
Prioritise; pick the ONE first metric

STEP 4
Prep a 2-minute readout

STEP 5
Debrief: 3 teams present

YOUR ROLE
You are Stripe's PM team. Volume is up, new-merchant activation is falling and authorisation dips in one region.

Deliver on one page:

- Top 5 questions, one per ladder rung
- Top 3 hypotheses, ranked
- The ONE metric to measure first + its guardrail

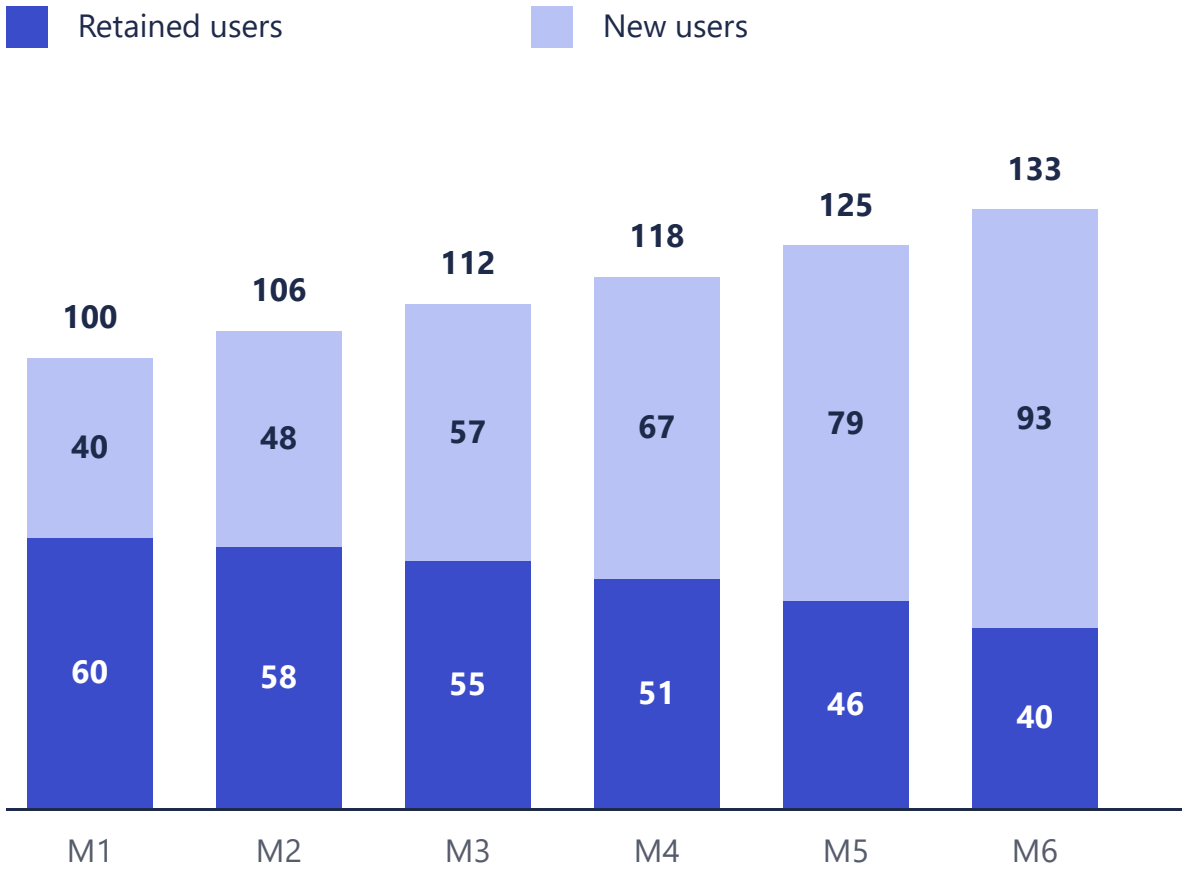
GROUP
Teams of 5; one plays "data"

FORMAT
Work as a team; then 3 teams present

DEBRIEF
What would change your first metric?

SESSION 7 · CLOSING THE LOOP

Growth can hide a leaky bucket



Illustrative: monthly active users (thousands).

+33%

total active users, M1 → M6

−33%

users retained from earlier months

Split growth into new vs existing. Read cohorts, not totals. Fix retention before buying more acquisition. **Your take-home question starts here.**

SESSION 7 · WRAP-UP

Six things to take with you

01

Start from strategy

A metric only helps if it traces back to a strategic bet.

02

Actionable beats vanity

Ratios, cohorts and per-user rates beat cumulative totals.

03

Spec it precisely

Definition, window, segment, owner, guardrail — or it gets argued.

04

One North Star, many inputs

Teams move inputs; the North Star moves with them.

05

Diagnose with structure

Real? Scope? Internal or external? Segment? Tree? Then test.

06

Retention tells the truth

Growth that doesn’t stay is just expensive traffic.



TAKE-HOME · OUTCLASS QUESTION

“When growth metrics look positive but retention dips, how would you decide what to measure and fix first?”

One page. Use the question ladder, name your North Star and one guardrail, and pick the first metric you’d fix — with your reasoning.

NEXT · SESSION 8: PRODUCT EXECUTION

Before next class

PRE-READ

Lenny’s Newsletter “How to write a great PRD”

Read with a pen: mark where the PRD informs, and where it commits.

REFLECTION

“Where in the product lifecycle does a PRD stop being a pitch and start being a contract with your team?”

